A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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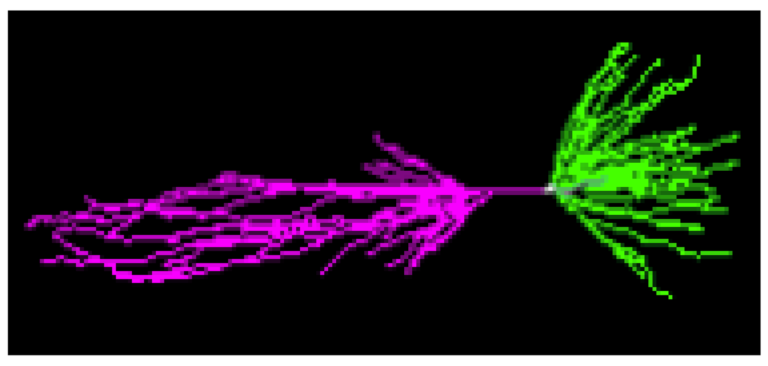
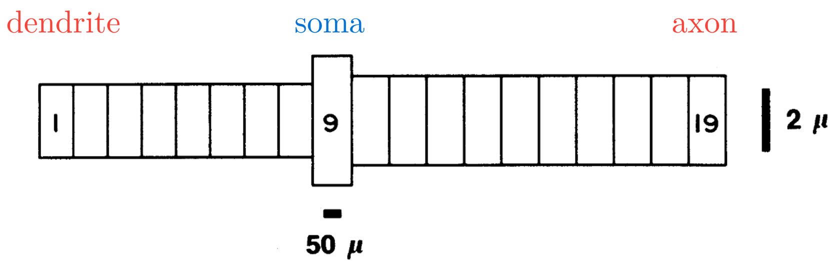
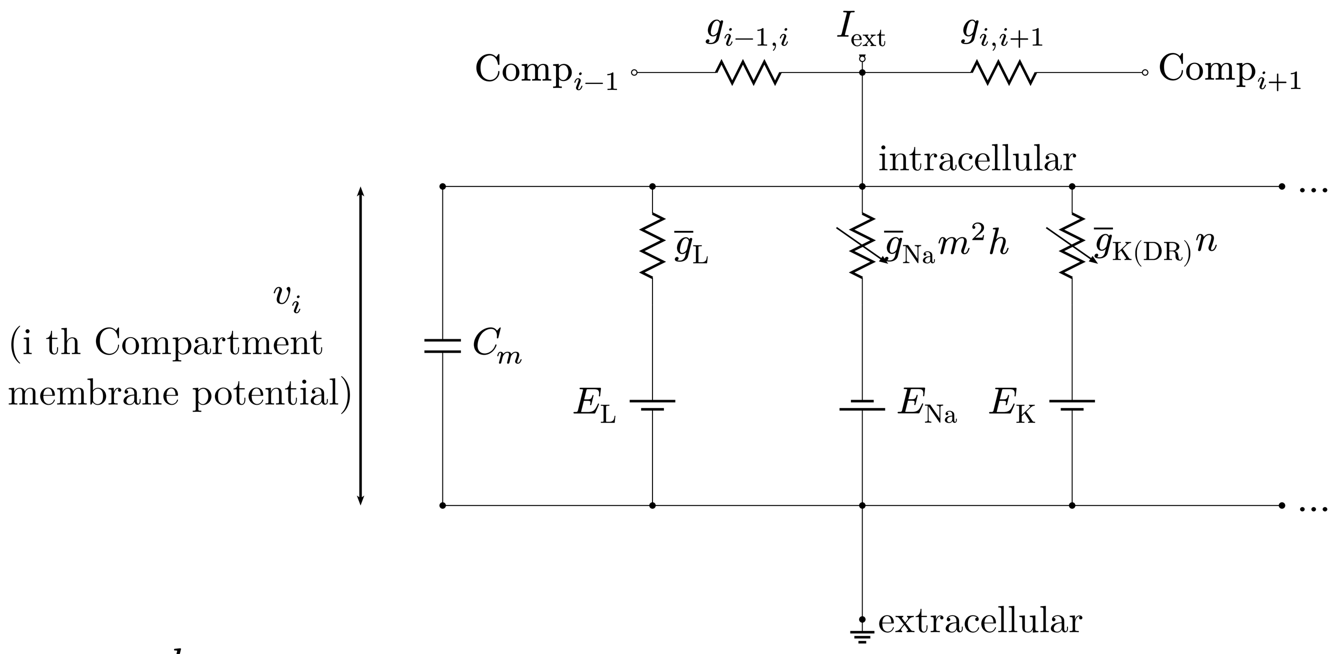
Introduction



Guinea pig CA3 pyramidal neuron[1]



Modeled as
19 compartments (comps) [2]



$$C_m \frac{dv}{dt} = -g_{\mathrm{leak}}(v - E_{\mathrm{leak}}) - \bar{g}_{\mathrm{Na}} m^2 h (v - E_{\mathrm{Na}}) - \bar{g}_{\mathrm{K}} n (v - E_{\mathrm{K}})$$
$$+ I_{\text{other ionic currents}} + I_{\mathrm{ext}}$$

The conductances depend on **gate variables**, which follow ODEs with the rate functions α, β:

$$I_{\mathrm{Na}} = \bar{g}_{\mathrm{Na}}\, m^2 h\, (v - E_{\mathrm{Na}})$$
$$\frac{dm}{dt} = \alpha_m(v)(1-m) - \beta_m(v) m$$
$$\alpha_m(v) = 0.32(13.1 - v)/(\exp((13.1 - v)/4) - 1)$$
$$\beta_m(v) = 0.28(v - 40.1)/(\exp((v - 40.1)/5) - 1)$$

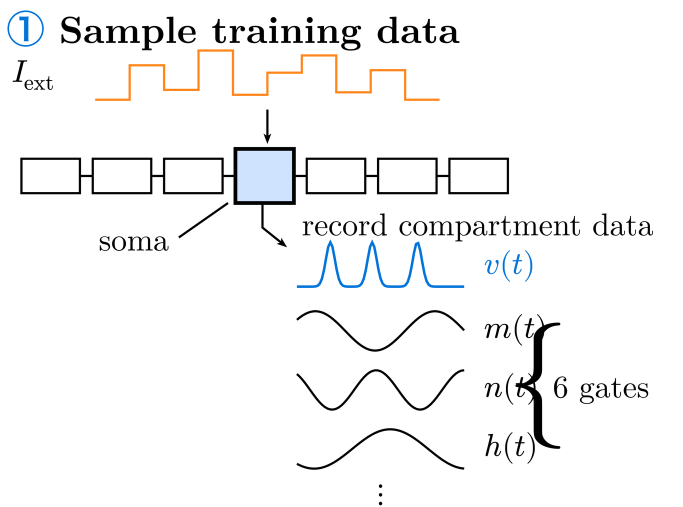
→ **10 gates and the potential v per comp**
→ **190 gates** for the 19 comps;
In large scale network simulations, the **large number of gate variables** becomes a **memory bottleneck**.

Purpose

Development of a surrogate model for the multi-compartment model, capable of reproducing the membrane potential response with fewer gate variables.

Methods

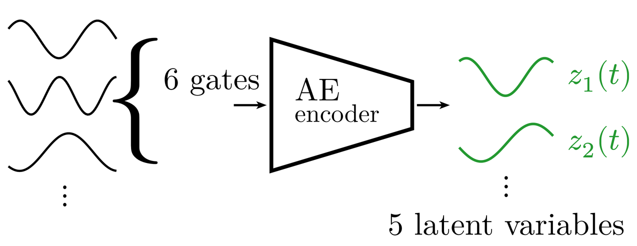
① Sample training data



Inject a **random pulse train** at the soma; record $v(t)$ and the **6 gates** $m(t), n(t), \ldots$.

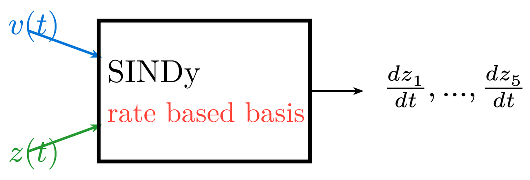
(The remaining gates, driven by Ca^{2+} dynamics, are left untouched.)

② Compress the 6 gates



The **6 gates** $m(t), n(t), \ldots$ are compressed to the **5 latent variables** $z_1(t), \ldots, z_5(t)$ by the encoder of an AutoEncoder (one hidden layer). $v(t)$ is **not** compressed.

③ Identify ODEs of the latent variables $z(t)$

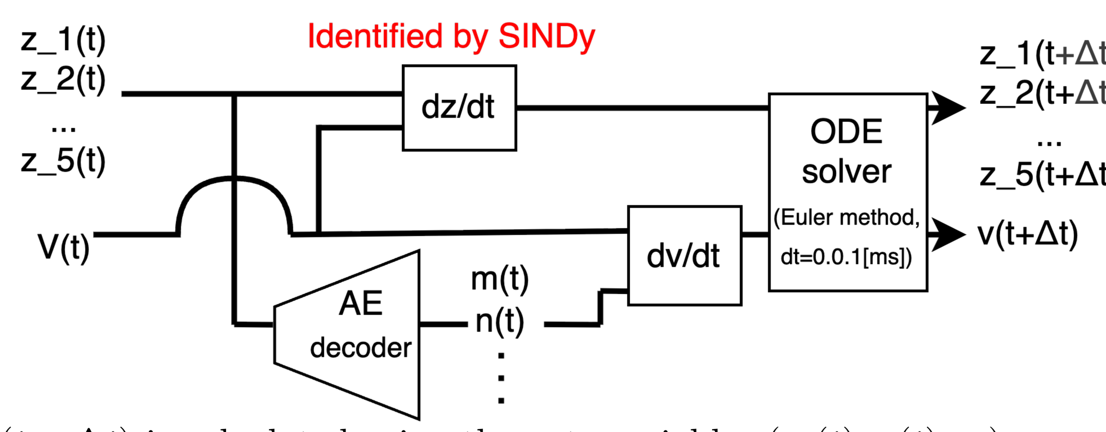


$$\frac{dz_1}{dt} = \xi_{11}\alpha_m(v) + \xi_{12}\beta_m(v) z_1 + \cdots$$
$$\frac{dz_2}{dt} = \xi_{21}\alpha_m(v) + \xi_{22}\beta_m(v) z_1 + \cdots$$
$$\vdots$$

To write the ODEs $\frac{dz}{dt}$ of the time series $z(t)$ as a linear sum of **basis functions**, the machine learning model **SINDy**[3] identifies the coefficients ξ_{ij}.

The basis functions are the rate functions $\alpha(v), \beta(v)$ of the gate variables, and their products with $z_1(t), \ldots, z_5(t)$.

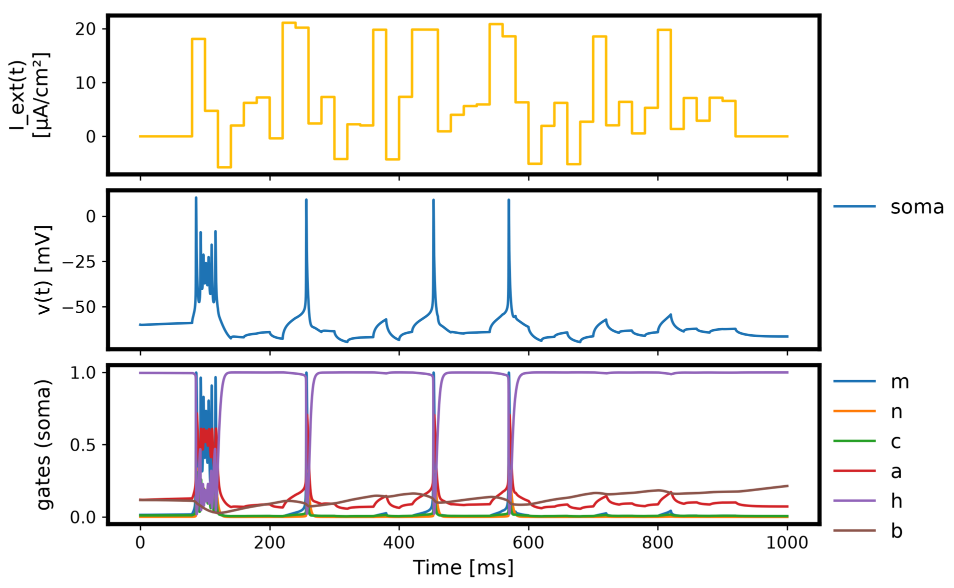
④ How to calculate the $v(t)$ with the compressed $z(t)$



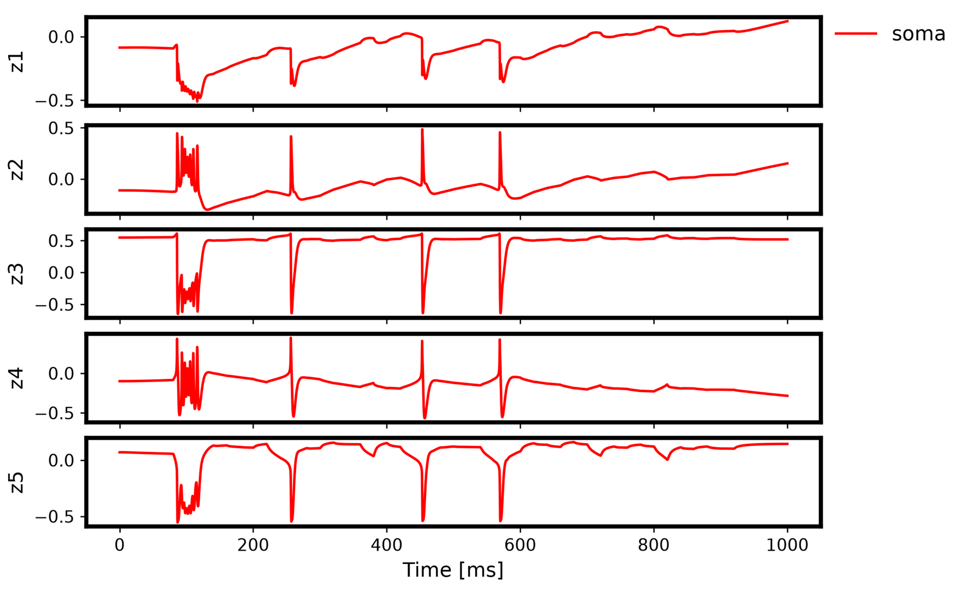
$v(t + \Delta t)$ is calculated using the gate variables $(m(t), n(t), \ldots)$ decoded by the **decoder** of the AutoEncoder from the latent variables $(z_1(t), z_2(t), \ldots)$.

Results and Discussion

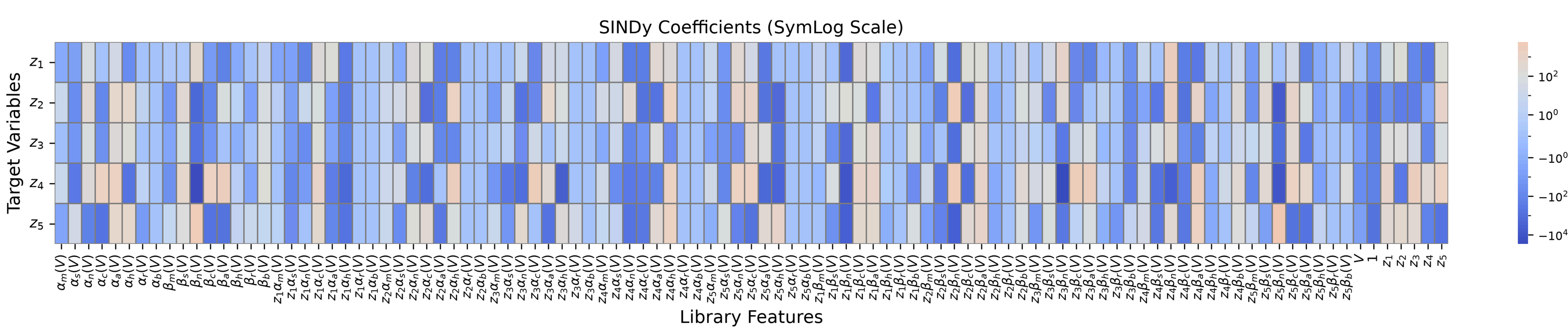
Training data to capture the gate dynamics



↓ **The 6 gates** are compressed into **5 latent variables**

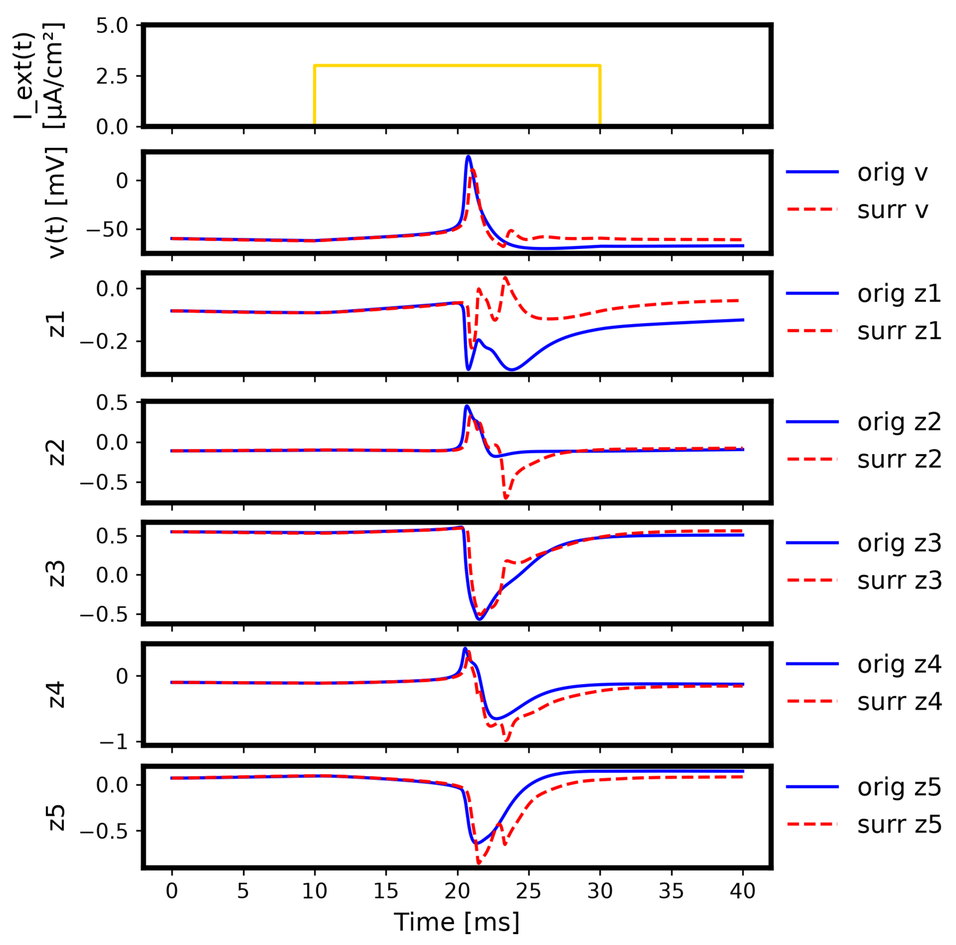


Identified equations of the latent variables



79.6 % of the coefficients of $\frac{dz}{dt}$ are non-zero.

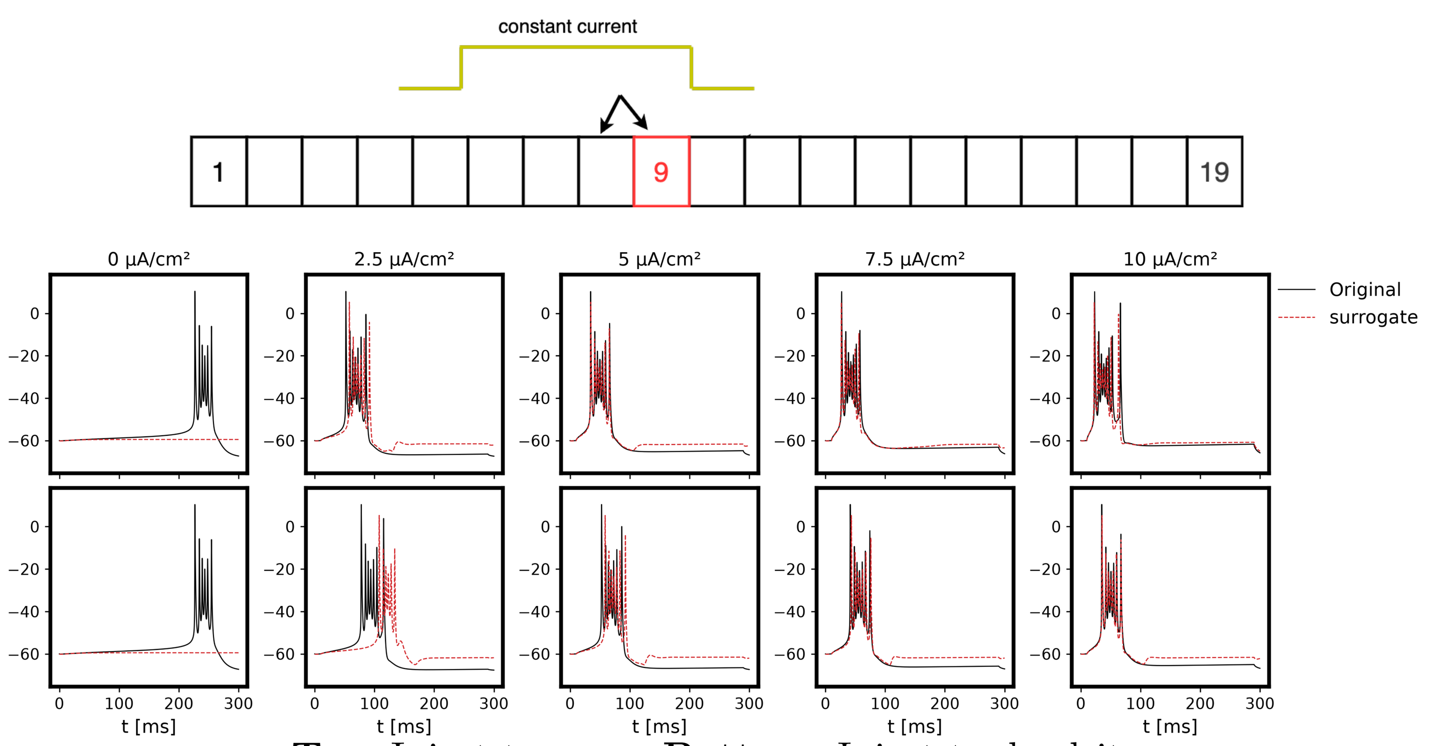
Reproducibility of a somatic surrogate model



- spike latency error **0.3 ms**, AHP timing gap **3.1 ms**.
- peak **13 mV low** (amplitude gap **12.9 mV**), rise / fall rate gap **21.0** / **10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**.

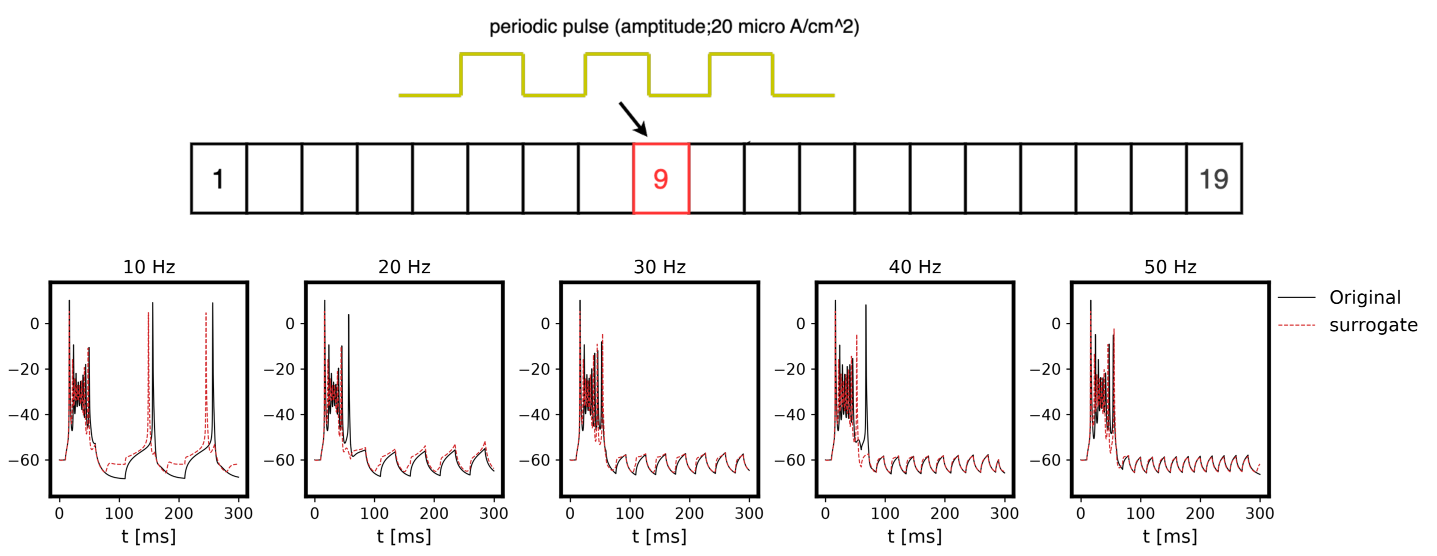
Reproducibility of a multi-compartment model in which the soma compartment was replaced by the somatic surrogate model

soma compartment (9th comp) is replaced with the surrogate model.



Top: Inject to soma. **Bottom**: Inject to dendrite.

- The surrogate model **reproduced the membrane potential well** for the constant currents.
- It did **not show the spontaneous firing** of the original model (left). The training data may not contain the spontaneous firing.



- For the pulse currents, the surrogate model **reproduced the membrane potential well** for $f \geq 30$ Hz.
- At 10 Hz, it fired about 10 ms earlier than the original model, and it did not show the spontaneous firing between the pulses.

Conclusion

- We simulated the multi-compartment model in which only the **soma compartment** was replaced by the surrogate model with **fewer gate variables (6 → 5)**.
- The surrogate model **reproduced the membrane potential well**, but it did not reproduce the dynamics that were **not contained in the training data**.

Future work

- Check the reproducibility when the number of gate variables is reduced further.
- Check the reproducibility of the **full surrogate model**, in which the compartments other than the soma are also replaced.

Code — github.com/MunechikaHaruki/SINDyNeuroSurrogate

[1] [Online]. Available: https://neuromorpho.org/neuron_info.jsp?neuron_name=L16a
[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.
[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.